

LESSON 5-6 • SESSION 2 • PRACTICE • FORM A

Reading Nets and Solids

Learning target – I can name the solid a net makes by reading the shapes in the net. • 6.G.A.4

REMEMBER

- **Prism:** 2 matching bases + rectangles. **Pyramid:** 1 base + triangles that meet at a point.
- The **base shape** names the solid: triangular, square, pentagonal, hexagonal.
- A **face** is flat, an **edge** is where faces meet, a **vertex** is a corner. Count them on the folded solid.

WORKED EXAMPLE

A net has 1 square and 4 triangles.

Step 1 $1 + 4 = 5$ faces.

Step 2 One square base + triangles: square pyramid.

Step 3 Edges $4 + 4 = 8$; vertices $4 + 1 = 5$.

Answer square pyramid: 5 faces, 8 edges, 5 vertices

WORDS **edge** (*arista*) • **vertex** (*vértice*) • **base** (*base*)

PRACTICE

show your work • label units

1. Name the solid each net makes.

a. 2 pentagons + 5 rectangles: _____

b. 4 triangles: _____

c. 6 identical squares: _____

d. 2 octagons + 8 rectangles: _____

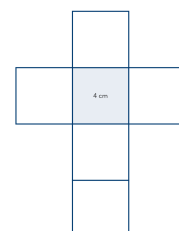
2. **Mystery Net.** A net shows 6 identical squares, each 4 cm on a side.

a. Name the solid. _____

b. Faces, edges, vertices _____

c. Area of one face _____ cm^2

d. Area of all the faces _____ cm^2



3. **Fold the Net.** A net has 2 rectangles 9 in by 5 in, 2 rectangles 9 in by 3 in, and 2 rectangles 5 in by 3 in.

Hint Add all six faces.

a. Name the solid. _____

b. Total area _____ in^2

Show your work.

4. **Game Box.** A net has 2 squares 12 in by 12 in and 4 rectangles 12 in by 3 in.

a. Name the solid. _____

b. Total area _____ in^2

Show your work.

5. **Cube Count.** A storage cube has 7 cm edges.

Hint Edges are the lines. Vertices are the corners.

a. Faces, edges, vertices _____

b. Area of one face _____ cm^2

Show your work.

6. **Ravi's Claim.** Ravi sees a net of 1 square and 4 triangles. Ravi says it folds into a triangular prism. Is Ravi right?

Hint Count the bases. One base means a pyramid.

Show your work.

Solid _____

7. **Ornament Net.** A net has 1 triangle base with 3 triangles around it. Each outer triangle has a base of 9 cm and a height of 8 cm.

Hint Use the triangle formula on one outer face.

Show your work.

Solid, area _____

8. **Three Nets.** Net 1: 2 triangles + 3 rectangles. Net 2: 1 square + 4 triangles. Net 3: 6 rectangles in 3 matching pairs.

- a. Name the solid each net makes. _____
b. Explain how you tell a prism from a pyramid. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Nadia's Work.** Nadia says a cube has 6 faces, 6 edges, and 6 vertices, because everything about a cube comes in sixes. Explain the mistake and give the correct counts.

TEST PRACTICE

choose the best answer

10. A prism is named for the shape of its –

A bases **B** tallest face **C** edges **D** vertices

11. A net with 1 hexagon and 6 triangles makes a –

A hexagonal prism **B** hexagonal pyramid
C cube **D** triangular pyramid

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

16 S	36 S	49 D	96 O	174 L	72 Q	432 I	87 Z
#2c		#2d	#3b	#4b	#5b	#7	

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____